

phase2-week4-day1-ingestion-pipeline-api

Ingestion Pipeline using API:

1. Health Check:

The screenshot shows the Postman interface with a workspace named "My Workspace". The left sidebar lists collections, including "RAG MongoDB - Data Ingestion" which contains a "1. Health Check" request. The main panel shows the details of this request: a GET method to the URL "http://localhost:3001/api/health". The "Body" tab is selected, showing "none". The "Response" tab is also visible, showing a successful response. The right sidebar displays "Variables in request", including "baseUrl" set to "http://localhost:3001" and "outputFile" set to "stories-1781976991154.j...".

Server is Running:

This screenshot is identical to the one above, showing the same Postman interface with the "1. Health Check" request. The "Response" tab is now active, displaying a successful response with a status of 200. The "Body" tab is still selected, showing "none". The right sidebar displays the same "Variables in request" as before.

The screenshot displays the Langflow web interface, which is divided into several panels:

- Left Panel (Navigation):** Contains a sidebar with a 'My Workspace' dropdown and a list of collections. The 'RAG MongoDB - Data Ingestion' collection is expanded, showing a list of endpoints. The 'GET 1. Health Check' endpoint is selected and highlighted.
- Top Panel (Search and Environment):** Features a search bar and a dropdown menu for selecting an environment, currently set to 'No environm...'.
- Center Panel (Request Editor):**
 - Method and URL:** A dropdown menu shows 'GET', and the URL is set to '({baseUrl}) /api/health'.
 - Body:** A dropdown menu shows 'none', and a message states 'This request does not have a body'.
 - Response:** The response status is '200 OK' with a time of '1.83 s' and a size of '312 B'. The response body is a JSON object:

```
{ "status": "OK", "message": "Server is running" }
```

.
- Right Panel (Variables):** Titled 'Variables in request', it lists variables used in the request:
 - baseUrl:** http://localhost:3001
 - outputFile:** stories-1781976991154.j...
 - jobId:** job-1781977168101-z8z6...
 - resources:** api/now/table
 - Local Vault:**
 - vault:groq_api_key_07hc:
 - vault:basic_auth_0n6h:

The screenshot displays the RAG Studio interface, which is used for building Retrieval-Augmented Generation (RAG) applications. The interface is divided into several sections:

- Left Sidebar (Collections):** Lists various data sources and workflows, including 'GenAI-May', 'GenAI-May-Home Work', 'Jenkins_Requests', 'JIRA', 'jiraTest Copy', 'Langflow_Workflows', and 'RAG MongoDB - Data Ingestion'. The 'RAG MongoDB - Data Ingestion' collection is expanded, showing a list of steps: 'GET 1. Health Check', 'POST 2. Upload Excel - Test Cases' (selected), 'POST 2. Upload Excel - User Stories', 'GET 3. List Available Files', 'POST 4. Create Embeddings - Batch (Test Cas...', and 'POST 4. Create Embeddings - Batch (User Stor...'.
- Main Workspace:** Displays the details of the selected workflow step, 'POST 2. Upload Excel - Test Cases'. It shows the URL 'POST {{baseUri}} /api/upload-excel' and a 'Send' button. Below this, there are tabs for 'Docs', 'Params', 'Authorization', 'Headers', 'Body' (selected), 'Scripts', and 'Settings'. The 'Body' tab shows a 'form-data' section with a table of key-value pairs:

Key	Value	Description	Bulk Edit
file	testcases.xlsx	Excel file containing test	
dataType	testcases	Type of data: 'testcases'	
sheetName	Testcases	Name of the Excel sheet	
- Right Panel:** Contains sections for 'Variables in request' (showing 'baseUrl' as 'http://localhost:3001'), 'All variables', 'Environment' (with a link to 'Select environment'), 'RAG MongoDB - Data Ingestion' (showing 'baseUrl' as 'http://localhost:3001', 'outputFile' as 'stories-1781976991154.j...', and 'jobId' as 'job-1781977168101-z8z6...'), 'Globals' (showing 'resources' as 'api/now/table'), and 'Local Vault' (showing 'vault:groq_api_key_07hc' and 'vault:basic_auth_On6h' as '.....').
- Bottom Bar:** Includes icons for 'Connect Git', 'Console', 'Terminal', and other utility tools.

API Response:

The screenshot displays the Postman interface for a REST client. The main workspace shows a POST request to the endpoint `{{baseUrl}}/api/upload-excel`. The request body is configured as form-data with three fields: `file` (type File, value `testcases.xlsx`), `dataType` (type Text, value `testcases`), and `sheetName` (type Text, value `Testcases`). The response status is 200 OK, and the body is a JSON object:

```
{
  "success": true,
  "message": "Test cases file converted successfully",
  "outputFile": "converted-1782024371428.json",
  "output": "\u26a0 Sheet \"Testcases\" not found. Using first sheet: \\\"testcases\\\" \u2705 Converted 6354 rows from \\\"testcases\\\" into E:/genai-classroom/Phase_2/RAG/ingestion/rag-mongo-demo-v9/src/data/converted-1782024371428.json"
}
```

The right sidebar shows the 'Variables in request' section with the following variables:

Variable	Value
baseUrl	http://localhost:3001
outputFile	converted-17820243714...
jobId	job-1781977168101-z8z6...
resources	api/now/table
vault:groq_api_key_07hc	
vault:basic_auth_0n6h	

The screenshot shows a file explorer window with the path `Phase_2 > RAG > ingestion > rag-mongo-demo-v9 > src > data`. The file list contains the following items:

Name	Date modified	Type	Size
testcases	20-06-2026 17:07	Microsoft Excel Work...	918 KB
userstories	20-06-2026 17:07	Microsoft Excel Work...	26 KB
converted-1782024371428	21-06-2026 12:16	JSON Source File	6,446 KB

3. Upload Excel – User Stories:

The screenshot shows the Postman interface for a collection named 'RAG MongoDB - Data Ingestion'. The selected item is '2. Upload Excel - User Stories'. The request is a POST to the endpoint `{{baseUrl}}/api/upload-excel`. The body is set to 'form-data' with the following fields:

Key	Value	Description	Bulk Edit
file	File	userstories.xlsx	Excel file containing user
dataType	Text	userstories	Type of data: 'testcases'
sheetName	Text	stories	Name of the Excel sheet

The right sidebar displays 'Variables in request' for the 'RAG MongoDB - Data Ingestion' environment:

Variable	Value
baseUrl	http://localhost:3001
outputFile	converted-17820243714...
jobId	job-1781977168101-z8z6...

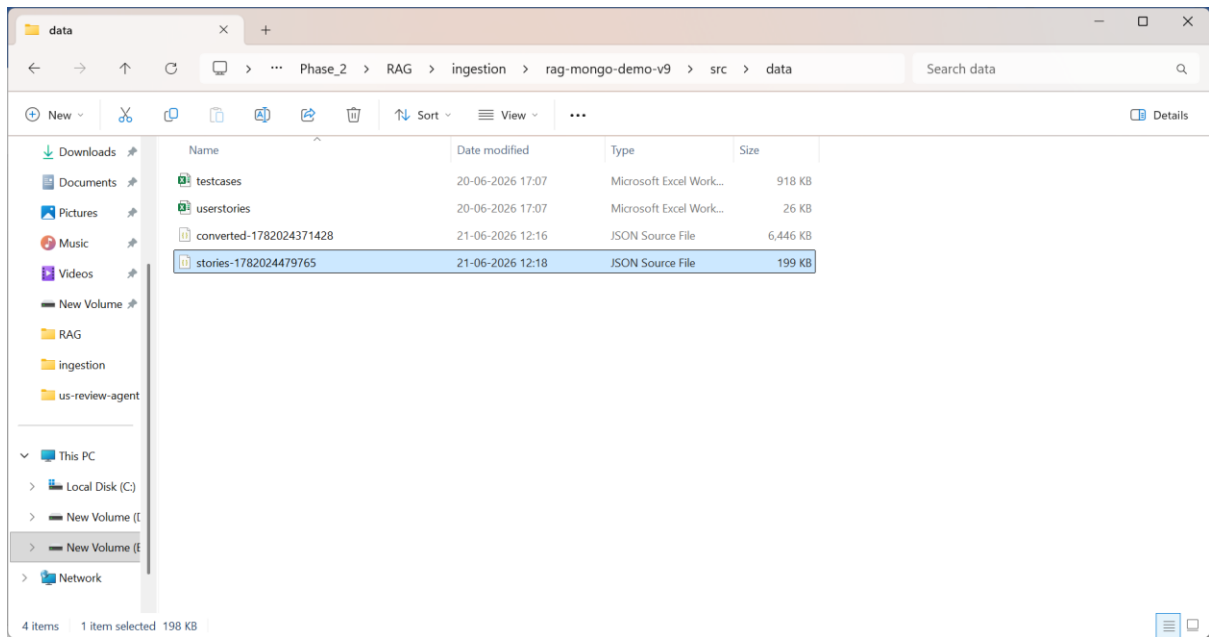
API Response:

The screenshot shows the Postman interface displaying the API response for the 'Upload Excel - User Stories' collection item. The response is a 200 OK status with a JSON body:

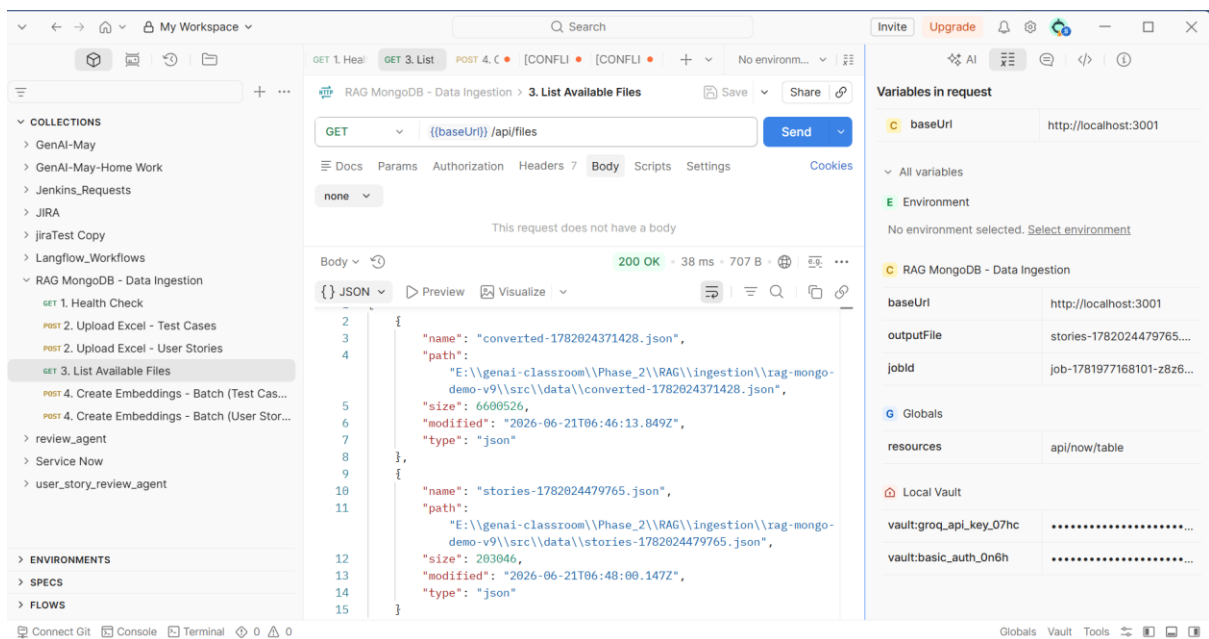
```
1 {
2   "success": true,
3   "message": "User stories file converted successfully",
4   "outputFile": "stories-1782024479765.json",
5   "output": "
    Converting 190 rows from sheet \"stories\"...
    Converted 190 user stories from \"stories\" into E:/
    genai-classroom/Phase_2/RAG/ingestion/rag-mongo-demo-v9/src/
    data/stories-1782024479765.json
    Filtered out: 0 empty
    rows
  "
6 }
```

The right sidebar displays 'Variables in request' for the 'RAG MongoDB - Data Ingestion' environment:

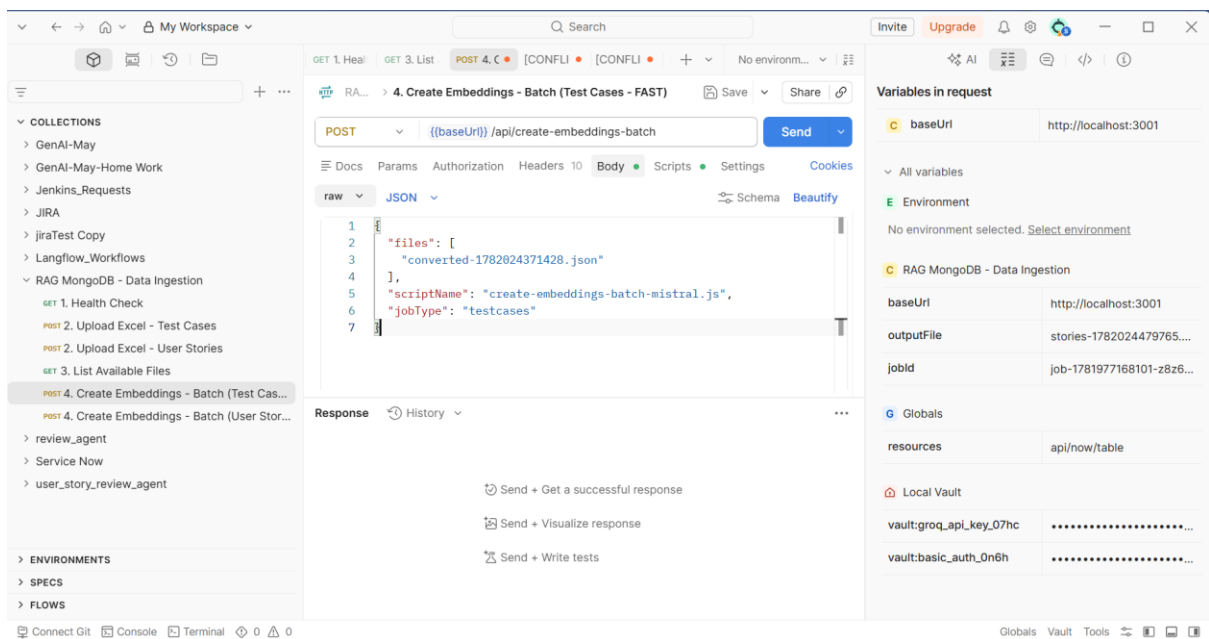
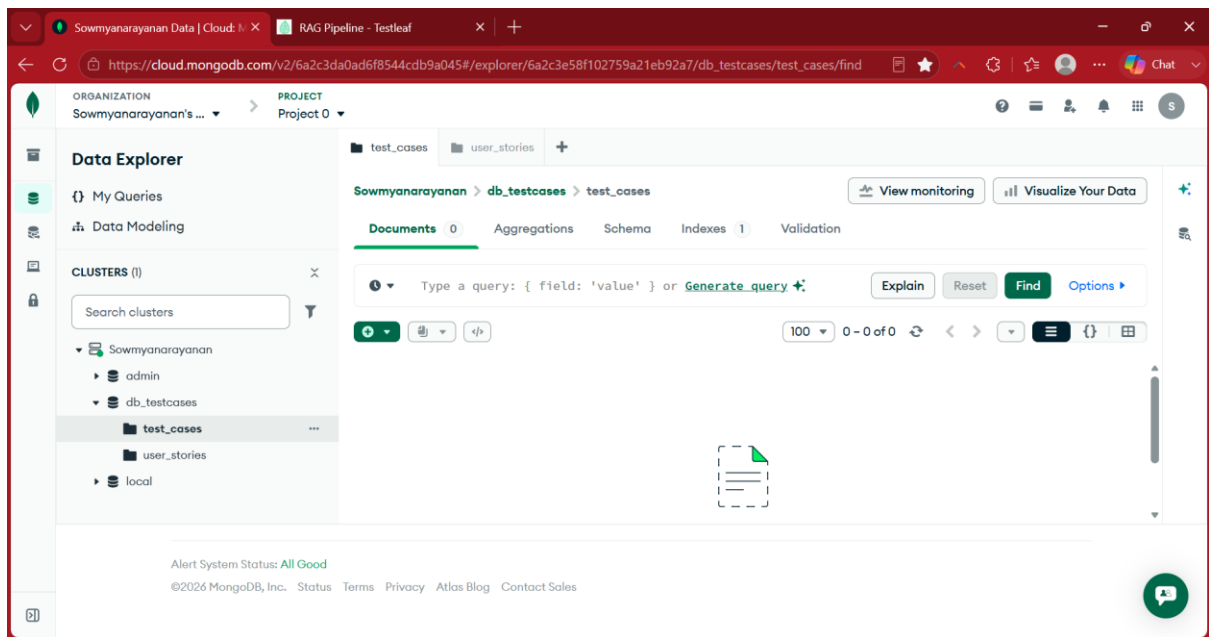
Variable	Value
baseUrl	http://localhost:3001
outputFile	stories-1782024479765...
jobId	job-1781977168101-z8z6...



4. List Available Files:



5. Create Embeddings – Test Cases



API response:

The screenshot shows an API client interface with a sidebar on the left listing collections and environments. The main panel displays a POST request to `POST 4. Create Embeddings - Batch (Test Cases - FAST)` with a body containing a JSON object. The response is a 200 OK status with a JSON body indicating success.

Request Body:

```
1 {
2   "files": [
3     "converted-1782024371428.json"
4   ],
5   "scriptName": "create-embeddings-batch-mistral.js",
6   "jobType": "testcases"
7 }
```

Response Body:

```
1 {
2   "success": true,
3   "jobId": "job-1782024682970-x7cyjnxvj",
4   "message": "Batch embedding creation started using
5     create-embeddings-batch-mistral.js",
6   "filesCount": 1,
7   "jobType": "testcases"
8 }
```

Variables in request:

Variable	Value
baseUrl	http://localhost:3001
outputFile	stories-1782024479765...
jobId	job-1782024682970-x7c...

Project Console:

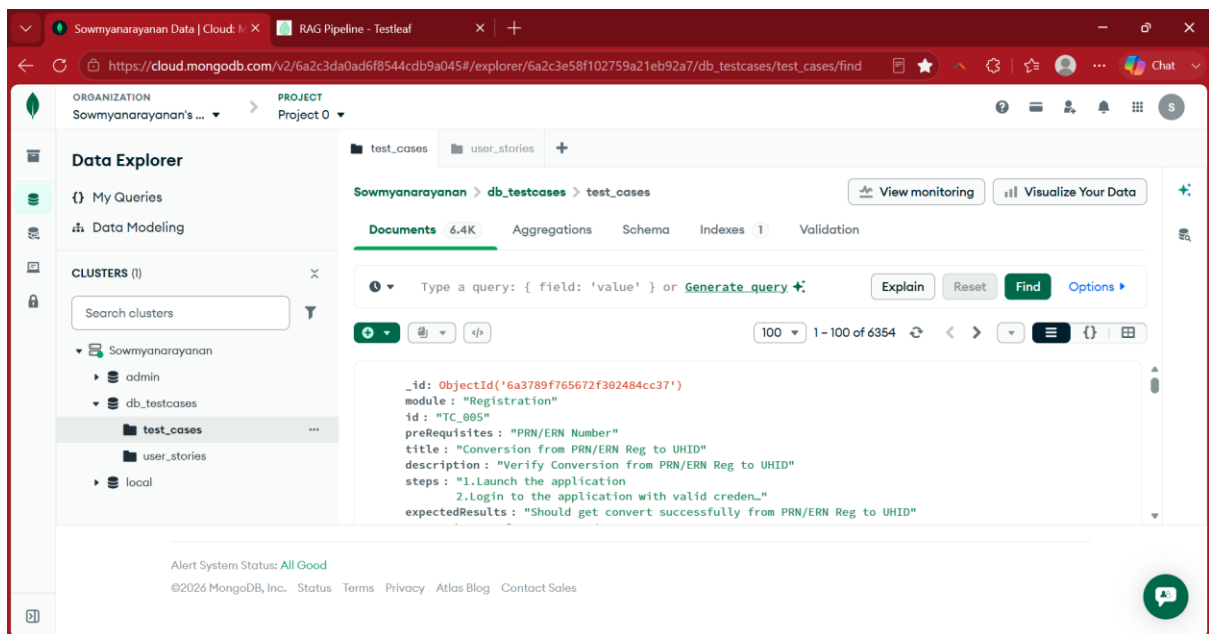
The screenshot shows a project console with a file explorer on the left and a terminal window on the right. The terminal displays the output of a script that creates embeddings for test cases.

Terminal Output:

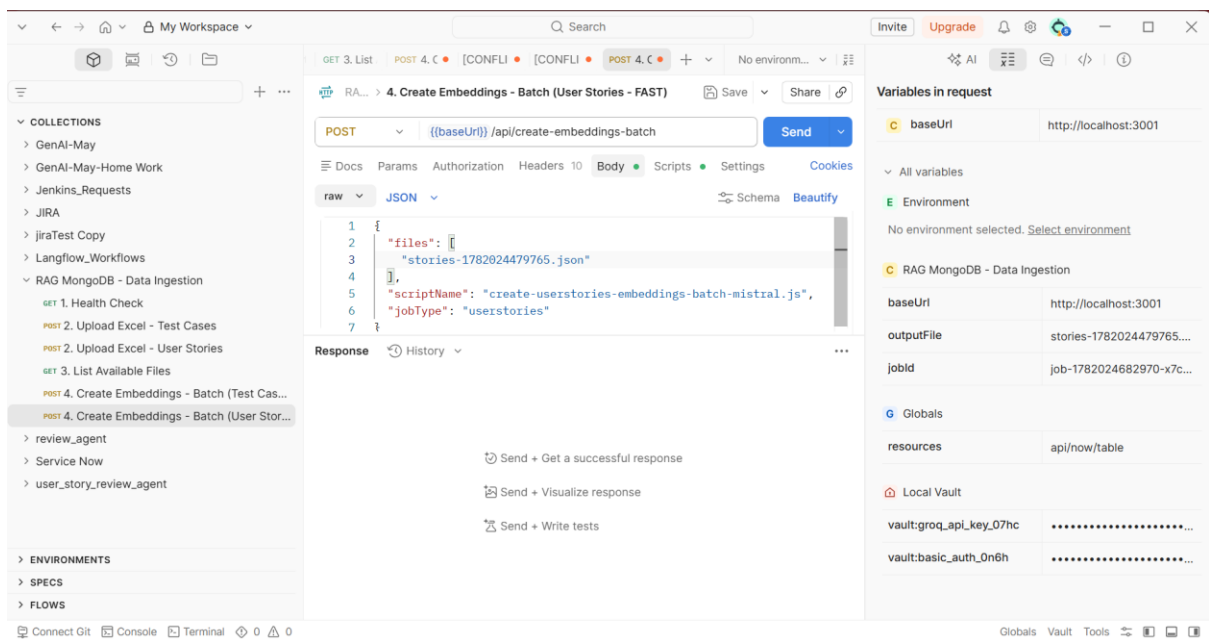
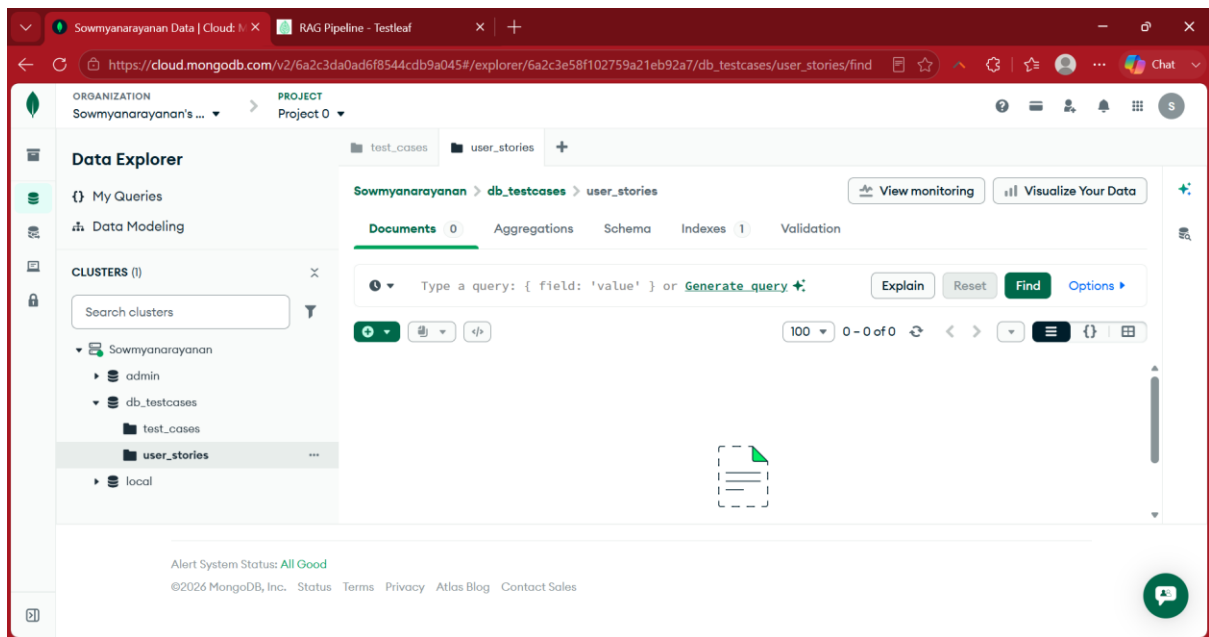
```
PS E:\genai-classroom\Phase_2\RAG\ingestion\rag-mongo-demo-v9> npm run dev
[0] [testcases] [Batch 1/128] Success! Cost: $0.000736 | Tokens: 7355
[0] [testcases] [Batch 2/128] Success! Cost: $0.000736 | Tokens: 7355
[0] [testcases] [Batch 3/128] Success! Cost: $0.000736 | Tokens: 7355
[0] [testcases] [Batch 4/128] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 5/128] Processing 50 testcases with Mistral AI...
[0] [testcases] [Batch 6/128] Processing 50 testcases with Mistral AI...
```

```
.env
1 MONGODB_URI="mongodb+srv://GenAIMay:2MwVBwFYdrzkV2tq@sowmyanarayanan_g1odcu.mongodb.net/?appName=Sowmyanarayanan"
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c514wj.mongodb.net/?retryWrites=true&w=majority"
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="test_cases"
7 VECTOR_INDEX_NAME="vector_index_testcases"
8 BM25_INDEX_NAME="testcases_bm25"
9
10 USER_STORIES_COLLECTION_NAME="user_stories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstories"
12 #USER_STORIES_BM25_INDEX_NAME="bm25_user_stories"
13
14 #USER_STORIES_COLLECTION_NAME="user_stories"
15 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"

PS E:\genai-classroom\Phase_2\RAG\ingestion\rag-mongo-demo-v9> npm run dev
[0] [testcases] ❌ Failed: 0
[0] 📊 Success Rate: 100.0%
[0] 💰 Total Cost: $0.117741
[0] [testcases] 📊 Total Tokens: 11,77,411
[0] 📊 Average Cost per Test Case: $0.0001853
[0] 📊 Average Tokens per Test Case: 185
[0] 💡 Cost Savings: ~90% cheaper than OpenAI embeddings
[0]
[0] ✅ Batch script completed successfully!
```



6. Create Embeddings – User Stories:



API Response:

The screenshot shows a REST client interface with a workspace named "My Workspace". The selected endpoint is "4. Create Embeddings - Batch (User Stories - FAST)" with a POST method. The request body is a JSON object:

```
1 {
2   "files": [
3     "stories-1782024479765.json"
4   ],
5   "scriptName": "create-userstories-embeddings-batch-mistral.js",
6   "jobType": "userstories"
7 }
```

The response status is 200 OK. The response body is a JSON object:

```
1 {
2   "success": true,
3   "jobId": "job-1782025142755-wdj9p5u5r",
4   "message": "Batch embedding creation started using
5     create-userstories-embeddings-batch-mistral.js",
6   "filesCount": 1,
7   "jobType": "userstories"
8 }
```

On the right, the "Variables in request" section shows:

- baseUri: http://localhost:3001
- outputFile: stories-1782024479765...
- jobId: job-1782025142755-wdj...
- resources: api/now/table
- Local Vault: vault:groq_api_key_07hc, vault:basic_auth_0n6h

Project Console:

The screenshot shows a project console with a file explorer on the left and a terminal on the right. The file explorer shows a project structure for "RAG-MONGO-DEMO-V9". The terminal shows the output of a script:

```
PS E:\genai-classroom\Phase_2\RAG\ingestion\rag-mongo-demo-v9> npm run dev
[0] [userstories] [Batch 1/4] Success! Cost: $0.000813 | Tokens: 8128
[0] [userstories] [Batch 2/4] Success! Cost: $0.000888 | Tokens: 8077
[0] [userstories] [Batch 3/4] Success! Cost: $0.000886 | Tokens: 8060
[0] [userstories] Waiting 1000ms before next batch group...
[0] [userstories] [Batch 4/4] Processing 40 user stories with Mistral AI...
[0] [userstories] [Batch 4/4] Success! Cost: $0.000641 | Tokens: 6413
```

VS Code interface showing the RAG-MONGO-DEMO-V9 project structure and the .env file configuration.

EXPLORER

- ✓ RAG-MONGO-DEMO-V9
 - client
 - node_modules
 - releases
 - server
 - src
 - config
 - data
 - converted-1782024371428.json
 - stories-1782024479765.json
 - testcases.xlsx
 - userstories.xlsx
 - scripts
 - uploads
 - .env
 - .env.example
 - .gitignore
 - package-lock.json
 - package.json
- OUTLINE
- TIMELINE

.env

```
1 MONGODB_URI="mongodb+srv://GenAIMay:2MwVBwFYdrzkV2tq@sowmyanarayanan.g1odcu.mongodb.net/?appName=Sowmyanarayanan"
2 #MONGODB_URI=mongodb+srv://sumitha:TestLeaf321@cluster0.c514wj.mongodb.net/?retryWrites=true&w=majority"
3 DB_NAME="db_testcases"
4 #DB_NAME="RagTest_cases"
5
6 COLLECTION_NAME="test_cases"
7 VECTOR_INDEX_NAME="vector_index_testcases"
8 BM25_INDEX_NAME="testcases_bm25"
9
10 USER_STORIES_COLLECTION_NAME="user_stories"
11 USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstories"
12 #USER_STORIES_BM25_INDEX_NAME="bm25_user_stories"
13
14 #USER_STORIES_COLLECTION_NAME="user_stories"
15 #USER_STORIES_VECTOR_INDEX_NAME="vector_index"
```

TERMINAL

```
PS E:\genai-classroom\Phase_2\RAG\ingestion\rag-mongo-demo-v9> npm run dev
[0] [userstories] Success Rate: 100.0%
[0] Total Cost: $0.003068
[0] [userstories] Total Tokens: 30,678
[0] [userstories] Average Cost per User Story: $0.00001615
[0] Average Tokens per User Story: 161
[0] Cost Savings: ~90% cheaper than OpenAI embeddings
[0] Batch script completed successfully!
```

